

Problem Session 1

Define notation in words

– $\in$

– $\notin$

– Ω

– $|$

– $\subset$

– $\cap$

– $\cup$

For the following bullets, practice reading / writing these as sentences + defining each set:

$$\Omega = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{x \in \Omega \mid x \text{ is even}\} = \{\dots\}$$

$$B = \{x \in \Omega \mid x \text{ is prime}\} = \{\dots\}$$

$$x \in \Omega \mid x \in A \cap B =$$

$$x \in \Omega \mid x \in A \cup B =$$

$$x \notin A \cup B =$$

$$A \cap (B \cup \{1, 9\}) =$$

$$x \in \Omega \mid (x \in A \cap B) \text{ or } (x \notin A) =$$

Set manipulation

Let A and B be two sets.

Using properties of set manipulation, show that...

$$A^c = (A^c \cap B) \cup (A^c \cap B^c) \text{ and } B^c = (A \cap B^c) \cup (A^c \cap B^c)$$

Please note: pictures are good for convincing yourself the statement is true, but are not generally considered mathematical proof.

Hint: recall the distributive property. When doing a proof, start with one side of the equation, and show that it equals the other side completely.

One-sided derivation = “Can I rewrite this expression step-by-step (ex. using algebra / set manipulation)?”

Complement of Intersections

Read this as a sentence:

$$- (\bigcap_n S_n)^c = \bigcup_n S_n^c$$

Show that: $(\bigcap_n S_n)^c = \bigcup_n S_n^c$

Two-sided proof = “Are these two sets actually the same?”

Intro to R/RStudio